Beam Alignment in IRS Aided mmWave Systems

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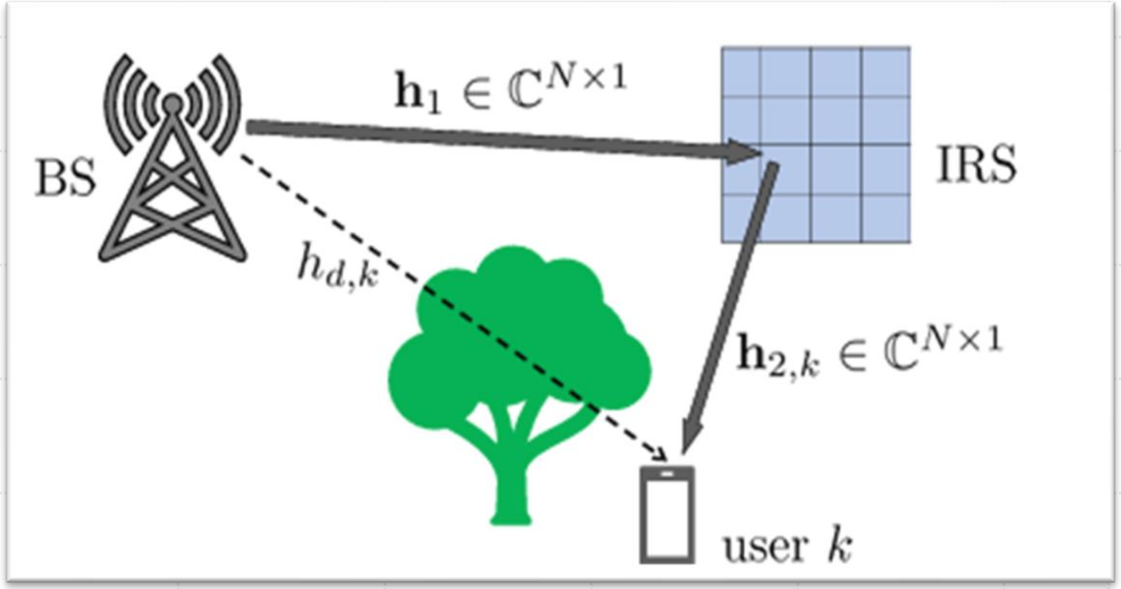
Intelligent Reflecting Surfaces

- **Intelligent Reflecting Surfaces (IRS):** Meta surfaces with passive elements
  - ❑ Alters the channel – "*reflect the signals in specific directions.*"
- Perks
  - S(I)NR
  - Energy efficiency
  - Coverage enhancement, etc.



Channel model:

$$h_{k,q}(t) = \sqrt{\beta_{r,k}}\mathbf{h}_{2,k}^H \boldsymbol{\Theta}_q(t)\mathbf{h}_1 + \sqrt{\beta_{d,k}}h_{d,k}.$$

The Benchmark (SISO systems)

Theorem 1 (Performance of the optimized IRS - beamforming configuration)

The rate obtained by the user k when the IRS is optimized to user k is

$$R_k^{BF} = \log_2\left(1 + \frac{P}{\sigma^2}\left|\sqrt{\beta_{r,k}}\sum_{n=1}^{N}|h_{1,n}||h_{2,k,n}| \times \exp(j\angle h_{d,k}) + \sqrt{\beta_{d,k}}h_{d,k}\right|^2\right).$$

It is achieved when (due to *Cauchy - Schwarz inequality*),

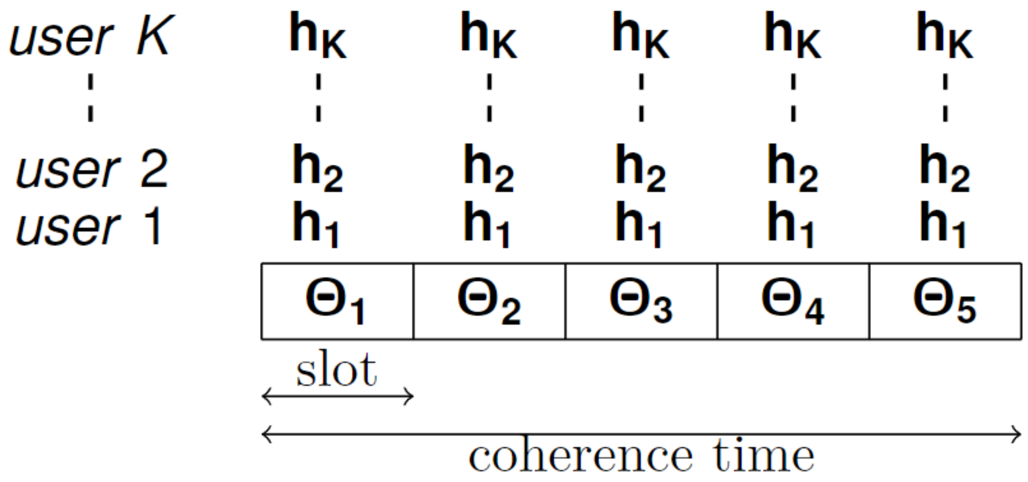
$$\theta_{n,k}^{BF} = \angle h_{d,k} - \angle(h_{1,n} + h_{2,k,n}), \qquad n = 1, \ldots, N.$$

- State-of-the art: "Three-fold overhead"
  - CSI estimation – complexity of O(N)
  - Expensive phase optimization
  - Transportation of phase angles

Can we **reap the benefits without** CSI estimation / phase optimization?

Prior Work[1,2]

Idea: Randomly configure the IRS & adopt opportunistic comm.



$$k^* = \arg\max \frac{R_k(t)}{T_k(t)}$$

Mean throughput

$$R_k(t) = \log_2\left(1 + \frac{P|h_{k,q}(t)|^2}{\sigma^2}\right)$$

Result: $\lim_{K\to\infty}\left(R^{(K)} - \frac{1}{K}\sum_k R_{BF}^{(k)}\right) = 0$

Random configuration ≈ beamforming (BF) configuration for **at least one UE**

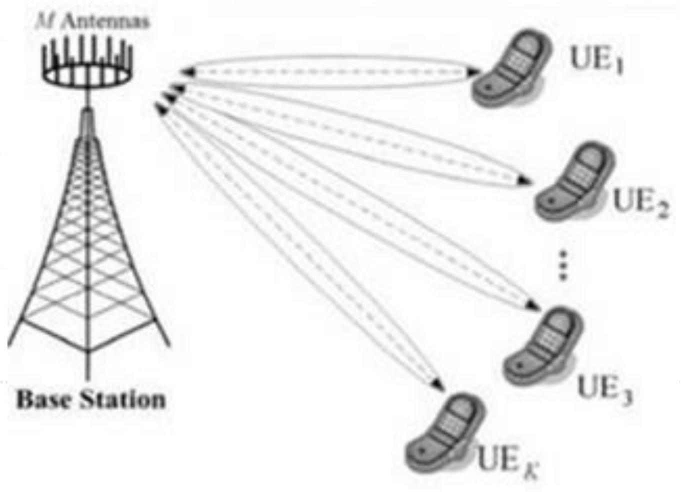
1) L. Yashvanth, and Chandra R. Murthy, *"Performance Analysis of IRS Assisted Opportunistic Communications"*, *IEEE Transactions on Signal Processing*

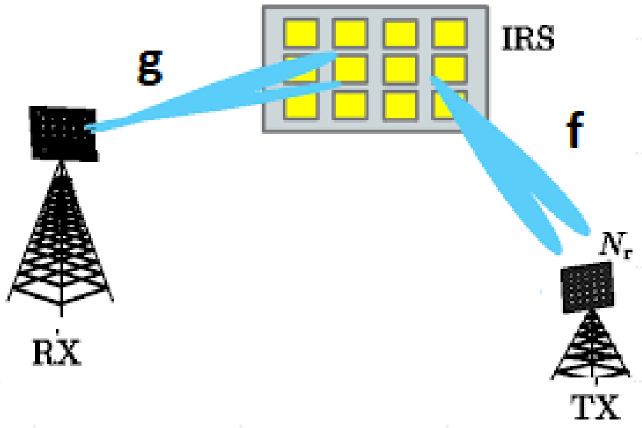
2) L. Yashvanth, and Chandra R. Murthy, *"Comparative Study of IRS Assisted Opp. Comm. Over i.i.d. and LoS Channels"*, *Proc. IEEE ICASSP, 2023*

This time...

- Randomized IRS aided Opp. comm. achieves optimality
  - Focus: system throughput – aids at least one UE at any time
- Can a randomized IRS help *a given* user?
  - Rapid BA via random beamforming
- **This talk:** Review of works on BA schemes in IRS systems
  - Attention to mmWave Systems

IRS in mmWave Systems





- Blockages are more! IRSs can help more!
  - IRS establishes link connectivity for mmWaves
- **Channel**: array steering response vector

$$\boldsymbol{a}(\theta) = \left[1, e^{-j\pi\sin\theta}, \ldots, e^{-j(M-1)\pi\sin\theta}\right]^T$$

- Modelling suitable for directional channels
  - Applicable in mmWave bands
- Model **f** and **g** via array steering vectors:

$$\boldsymbol{f} = \sum_{i=1}^{L_1} \alpha_i \boldsymbol{a}(\phi_i) \quad \boldsymbol{g} = \sum_{j=1}^{L_2} \beta_i \boldsymbol{a}(\psi_i)$$

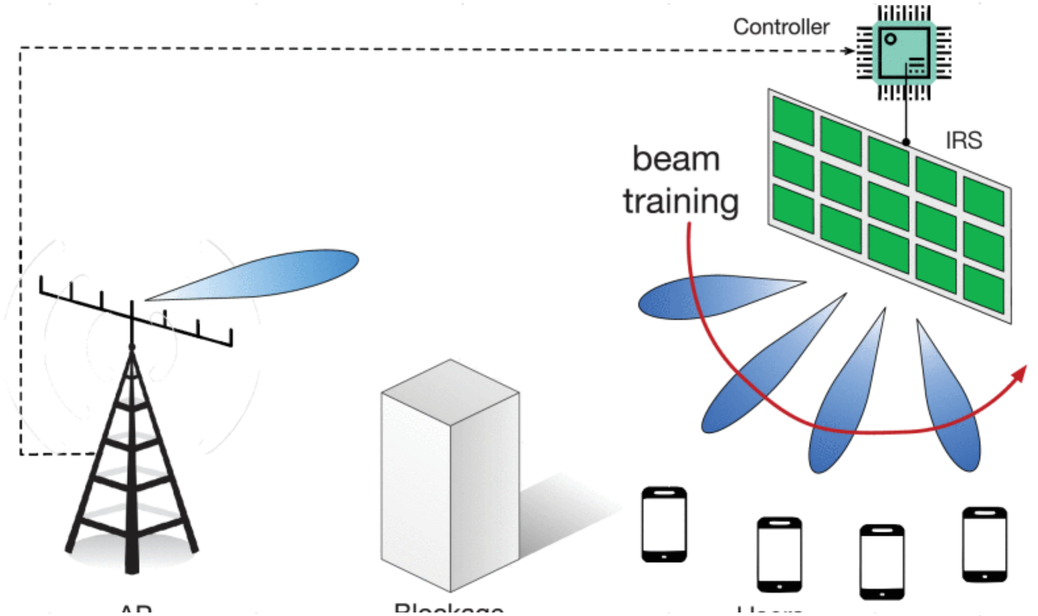
An IRS can favour sub-6 GHz too through rich-scattering

On the optimality of the IRS via Beam Training

- ✓ Optimal IRS configuration – depends on the full CSI
- ✓ High overhead with full CSI: scales with # BS antennas, # IRS elements
- ✓ ***Fact***: Existence of strong LoS *sometimes* (~ 20 dB stronger than NLoS paths)
- ✓ ***Idea***: Identify the dominant path, & **align** the IRS with the **dominant** path
- ✓ ***Optimal phase shift*** :- *Like* **Analog beamforming**: $\boldsymbol{\theta}^* = \mathrm{argmax}_{\boldsymbol{\theta}} \, |\mathbf{a}(\phi)\boldsymbol{\theta}|^2$
- ✓ **Circumvent** CSI estimation via beam training

Beam Training (or) Beam Alignment

- **Principle:** Alignment of txtr. and rcvr. beams along the dominant path
- **Utility:** Identification of connectivity links for *initial access in 5G*
- Both BS – IRS, and IRS – UE BA are required
    - IRS is passive, so BA is challenging
- **Naïve** implementation:
    - Beam – sounding at different angles
    - UE reports the "best beam" (via received SNR)



- Explore better & efficient techniques in coming slides

Parameters for BA in IRS - mmWave S/S

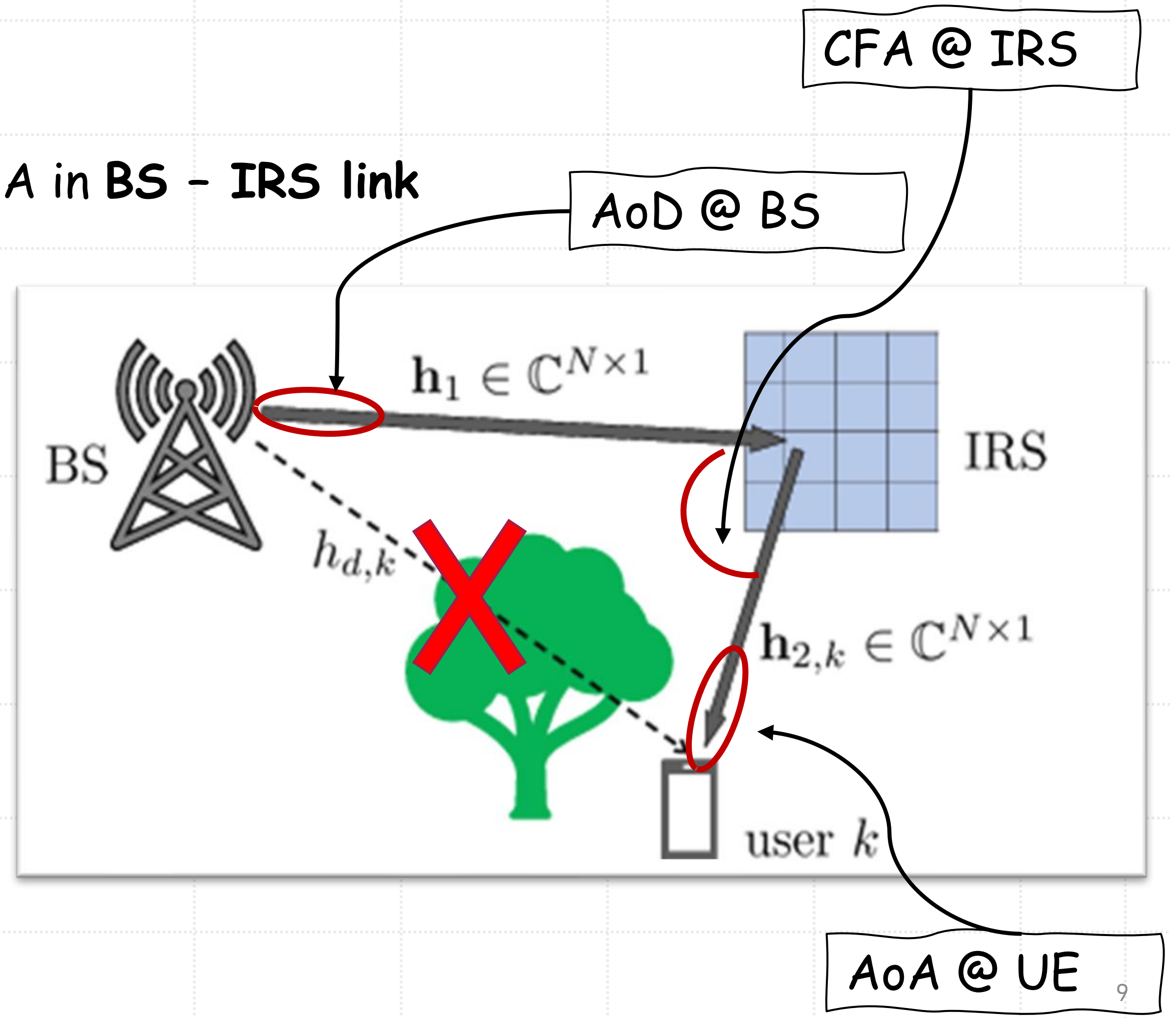
- Four – parameter based BA:
  - Angle – of – Departure (AoD) @ BS,
  - Angle – of – Arrival (AoA) @ IRS:

  } BA in **BS – IRS link**

  - AoD @ IRS;
  - AoA @ UE

  } BA in **IRS – UE link**

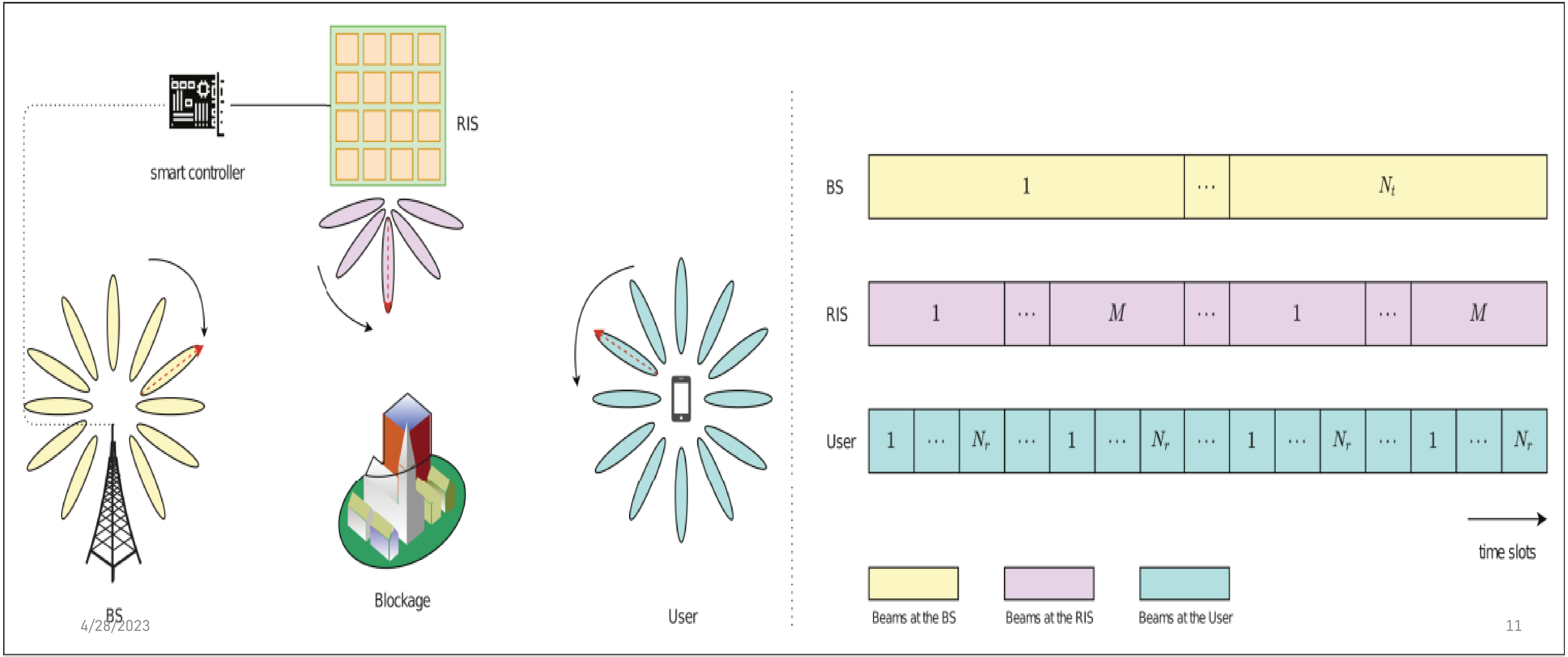
- **Simplified** parameter based BA:
  - AoD @ BS
  - cascaded reflective angle (CFA) @ IRS
  - AoA @ UE



Approach 1: Exhaustive Search

- Pre-defined code books at BS, IRS, and UE: $\mathcal{F}_B, \mathcal{F}_R, \mathcal{F}_U$
- Code books consist of array steering vectors for AoD, CFA, and AoA
- Beam tuple selection: $\{\theta_1^*, \theta_2^*, \theta_3^*\} = \mathrm{argmax}\ \mathrm{SNR}(\theta_1, \theta_2, \theta_3)$
  - Search count: $|\mathcal{F}_B| \times |\mathcal{F}_R| \times |\mathcal{F}_U|$
- Merit:
  - Larger beamforming gain if a beam exactly aligns with the dominant path even at low SNR
- Demerit:
  - High training complexity: $\mathcal{O}\left(N_t M N_r\right)$.
    - E.g $N_t = N_r = 32, M = 256$, then we need 262,144 training samples! – massive delay in initial access

Illustration of Exhaustive approach



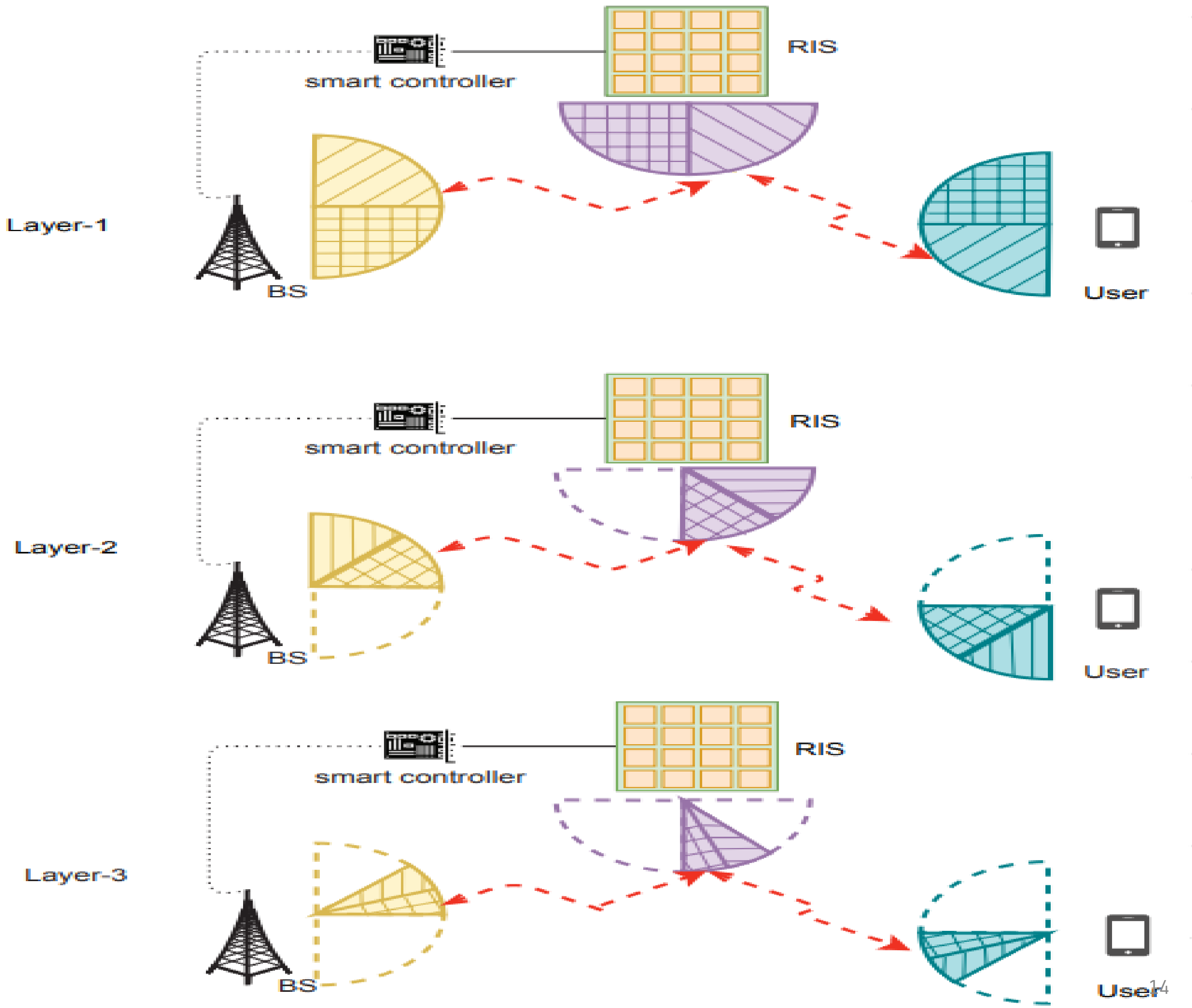
Approach 2: Hierarchical search

- It's multi-stage & interactive BA strategy
- How does it work?
  - The beam-width of the beam reduces in every layer
  - The spatial range of sounding depends on best beam in the previous stage
    - Adaptation of codebooks at BS, IRS & UE
  - Demerit:
    1. Feedback to BS and IRS in every stage to inform the "***direction***" of search
       - Exacerbated if the # of antennas / IRS elements increases
    2. Error propagation @ low SNR: Early wide beams could miss the dominant path

Hierarchical search contd…

- Merit: Significant reduction in the training overhead
  - E.g.: Binary-tree based progression and elimination: $\mathcal{O}\left(\log_2 \max\{N_t.N_r, M\}\right)$
  - Revisiting the example: $N_t = N_r = 32, M = 256$ - we need 8 layers to complete the search.
    - First 5 stages: there are $2^3 = 8$ beam tuples in each stage. So total number of tuples = 5x4 = 40
    - Next 3 stages: 2x3 = 6 beams.
    - Total of 46 beam tuples (or) training samples only! ☺

Illustration of Hierarchical search



Approach 3: Multidirectional Beam Search

- Leverage sparsity in mmWave channels
- Idea: Each node forms "*multiple*" beams independently

Beam Generation @ BS:

- Fully-connected hybrid A-D architecture, create R_{BS} beams = # RF chains
- These *beams* will be R_{BS} columns of the $N_t \times N_t$ DFT matrix

Beam Generation @ UE: Similar to BS: Form R_{UE} simultaneous beams from $N_r \times N_r$ DFT matrix

Beam Generation @ IRS:

- Reflection vector to be *linear combination* of Q columns of DFT matrix
  - Amplitude of each IRS element has to be individually controlled.
  - Obtain the nearest phase shifter via constrained optimization

Best BA Strategy

- 3D-angular space as a 3D-cube: uniformly divided into MN_tN_r small blocks
  - **Block**: A single beam tuple – potential cascaded BS-IRS-UE path
- Formation of **bins**:
  - Divide the space into $S = \dfrac{MN_tN_r}{R_{BS}R_{UE}Q}$ bins
- The bin with the dominant path (block) gives the best SNR: *"best"* bin, $\mathcal{B}_l$

'S' iterations *are* *necessary* *to scan the* *complete beam space!*

a.k.a. ***batch mode scanning***

How to identify the correct block?

- Repeat the entire search for L rounds (so total ***SL*** time complexity)
- Randomize the block allocation for all bins in every round
- Collect the common block indices among all the best bins!

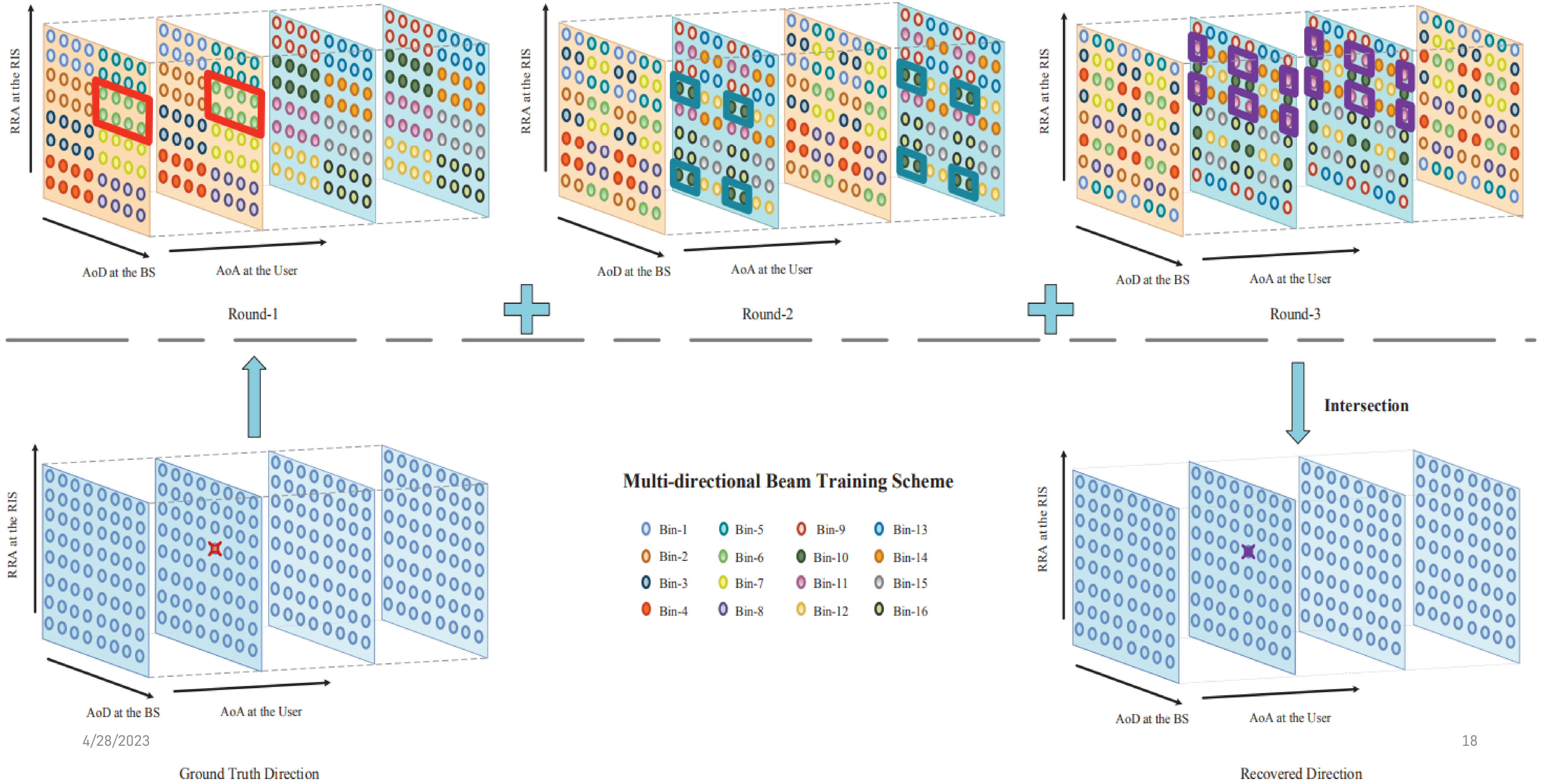
$$\{i\} \subset \cap_{l=1}^{L} \mathcal{B}_l$$

- **Premise**: No two blocks lie in the intersection of all the best bins

$$\lim_{L \to \infty} \left| \cap_{l=1}^{L} \mathcal{B}_l \right| = 1$$

- The best block or beam is identified: $\boxed{i^* \in \lim_{L \to \infty} \cap_{l=1}^{L} \mathcal{B}_l}$

Illustration ($M = 8, N_t = 8, N_r = 4, R_{BS} = 4, R_{UE} = 2, Q = 2$)



MDBS is moderately complex!

- What should be the minimum L to achieve a given probability of detection?
  - Ans: $\mathcal{O}\left(\log_2 \max\{N_t.N_r, M\}\right)$
- Comparison:

Parameter setting	Training Overhead	Perfect BA probability
$R_{BS}=8, R_{UE}=4, Q=16, L=4$	$T=2048$	95.76%
$R_{BS}=4, R_{UE}=4, Q=32, L=4$	$T=2048$	95.76%
$R_{BS}=4, R_{UE}=4, Q=32, L=5$	$T=2560$	99.64%

Training overhead vs detection probability for perfect BA.
Exhaustive protocol needs 262, 144 time slots for the search

Merits over the first two approaches

- NO multiple rounds of interaction between BS/IRS and UE
  - Less feedback overhead
  - Easily extendable to multi-UE scenarios
- Robust to low SNR
  - At any time narrow beams are used unlike hierarchical scheme

Summarizing and contrasting the schemes

Features	Algorithms		
	Exhaustive search	Hierarchical search	MDBS
Training overhead	$N_t N_r M$ Extremely high overhead	$C^3 \log_C \max\{N_t, N_r, M\}$ Low overhead	$\frac{N_t N_r M L}{N_{BS} N_{UE} Q}$ Moderately low overhead
Robustness against noise	Extremely high	Susceptible	High
Frequent feedback	Not required	Required	Not required
Extension to MU scenarios	Straightforward	Requires careful co-ordination	Straightforward

*C denotes the number of beams in each layer.

Next Steps

- The described protocol, minimizes the CSI estim. error with NO overhead constraint
- If overheads increases, it becomes comparable to optimization of IRS
- Pose an optimization problem:
  1. minimizes the training time
  2. *subject to*, a tolerable CSI estimation error
  3. via random passive beamforming
  - Optimal Random phase pattern design
  - Sub-array approach and optimal partitioning scheme
  - Demonstrate that a randomized IRS can help **any given user** with low complexity

References

- P. Wang, J. Fang, W. Zhang, Z. Chen, H. Li and W. Zhang, "***Beam Training and Alignment for RIS-Assisted Millimeter-Wave Systems: State of the Art and Beyond***," in IEEE Wireless Communications Magazine. December 2022
- P. Wang, J. Fang, W. Zhang and H. Li, "***Fast Beam Training and Alignment for IRS-Assisted Millimeter Wave/Terahertz Systems***," in IEEE Trans. on Wireless Communications, April 2022

Thank You!!

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